

FA23-BCE-113
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MSI-Lab # 4

Task 1: Write an assembly program to count the number of 1's using LOOP

Write an assembly language program that counts the number of '1's in a byte residing in CL register. Store the counted number in DH register.

```
MOV CL, 10110101B
XOR DH, DH          ; 1's count
MOV AL, CL           ; Copy byte to AL
MOV BL, 8            ; loop counter , no of bits

LOOP_START:
    AND AL, 1        ; Check LSB
    ADD DH, AL        ; Final value of 1's count
    SHR CL, 1        ; Shift right original byte for next bit
    MOV AL, CL        ; Move shifted value to AL again
    DEC BL            ; if zero flag is not zero jump
    JNZ LOOP_START   ; Repeat until all bits checked
```

registers		H	L
AX		00	00
BX		00	00
CX		00	00
DX		05	00

Description: Above code is well commented at every step.

Task 2: To identify an odd number

Write an assembly language program, that check the contents of AH register and place a 1 in BL register if the number is Odd or a 0 otherwise.

```
MOV AH , 22    ; number to check even
MOV BL , AH    ; move ah to bl
AND BL , 1     ; and bl with 1
```

EX	00	01
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Description: Number to check even is in ah, we move it to bl, AND with 1, if LSB is 1 (odd) then bl is 1 else 0(even)

Task 3: Write an assembly language program using subroutines

Write an assembly language program to calculate the average of four 8-bit numbers that uses a subroutine named 'ADDITION'. The numbers are stored in AL, AH, BL and BH registers and their average value should be left in CX. For adding the numbers, you are required to make use a subroutine "ADDITION"

```
MOV AL , 3
MOV AH , 3
MOV BL , 3
MOV BH , 3
CALL addavg
XOR AH , AH
RET
addavg:
ADD AL , AH
```

```
ADD AL , BL
ADD AL , BH ; al = sum
SHR AL , 2 ; al = avg
RET
```

